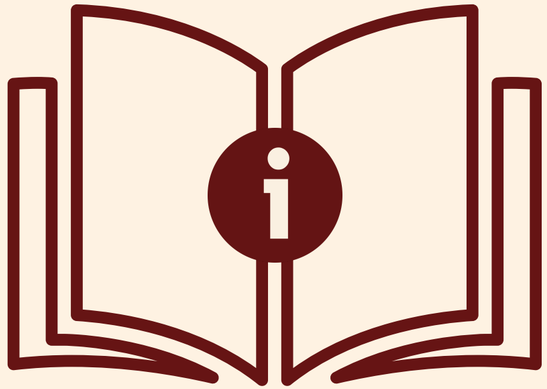
UNIVERSITÀ DEGLI STUDI DI MILANO BICOCCA

MILAN RESTAURANT RATINGS ACROSS ONLINE PLATFORMS

Data Management Project

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Introduction



Goal and Research Questions

- **Objective**: build a unified dataset to compare online restaurant ratings, characteristics and data quality
- **Platforms:** Google Maps, Tripadvisor and TheFork.
- **Location**: Milan







Today we will answer:

RQ1. Are some platforms systematically optimistic/pessimistic?

RQ2. Does rating consistency or level vary by cuisine?

RQ3. Does location affect data completeness?

Technologies and tools used

One of the project's focuses was to enable high collaboration and reproducibility within the team. That meant that choosing the appropriate technology and tools stack is crucial for the quality of the project:





Code & Reproducibility



Coding & Data Labeling



Coding & Code Review



Collaboration & Versioning





Web Scraping







Infrastructure & Databases





Analytics

ELT Paradigm



Data Acquisition (Extract)

Google Places API

- Official Google Places API: tiled Nearby Search + Place Details.
- Overlapping circular tiles and dense-neighbourhood anchors improved coverage in high-density areas.
- Checkpointed execution and place_id deduplication made the extraction safely repeatable.
- **Raw output: 10,808 place records, JSONL, 249 MB.**

Tripadvisor — Web Scraping

- Playwright scraper over listing pages and restaurant detail pages.
- Persistent browser sessions and checkpoints handled anti-bot friction and interruptions.
- Captured ratings, reviews, cuisines, prices, opening hours and contact fields; coordinates were added later.
- **Raw output: 7,539 restaurant records, JSON, 49 MB.**

TheFork — Web Scraping

- Custom Python scraper with listing discovery followed by detail-page enrichment.
- Proxy rotation and GraphQL/CDP recovery completed records blocked during detail navigation.
- Captured coordinates, ratings, reviews, discounts, cuisines, hours and price signals.
- **Raw output: 1,344 restaurant records, JSON, 11 MB.**

Data Loading



raw files

MongoDB.

Why MongoDB

- Stores heterogeneous data without requiring a fixed schema: Google Places, Tripadvisor and TheFork do not expose the same attributes and do not organize restaurant information according to a shared schema.
- Supports JSON-based documents with different attributes and levels of completeness.
- Preserves the original structure of each source before transformation

Configuration adopted

- Single loader used for all platforms.
- Natural key used as MongoDB document identifier.
- Google Places dataset streamed line-by-line. Tripadvisor and TheFork loaded directly from JSON arrays.

Source	Format	Natural Key
Google Places	JSON Lines	place_id
Tripadvisor	JSON Array	source_url
TheFork	JSON Array	source_id

Data Quality Assessment

Results

	Google	Tripadvisor	TheFork
Completeness	93.79%	84.34%	76.15%
Critical	98.15%	99.78%	98.88%,
Validity	100%	99.99%	100%
Spatial Readiness	100%	0%	100%
Timeliness	67.23%	43.75%	95.83%
Reliability	79.50%	53.10%	88.84%
Uniqueness	99.85%	99.97%	100%
Quality Score	93.16%	72.28%	97.63%

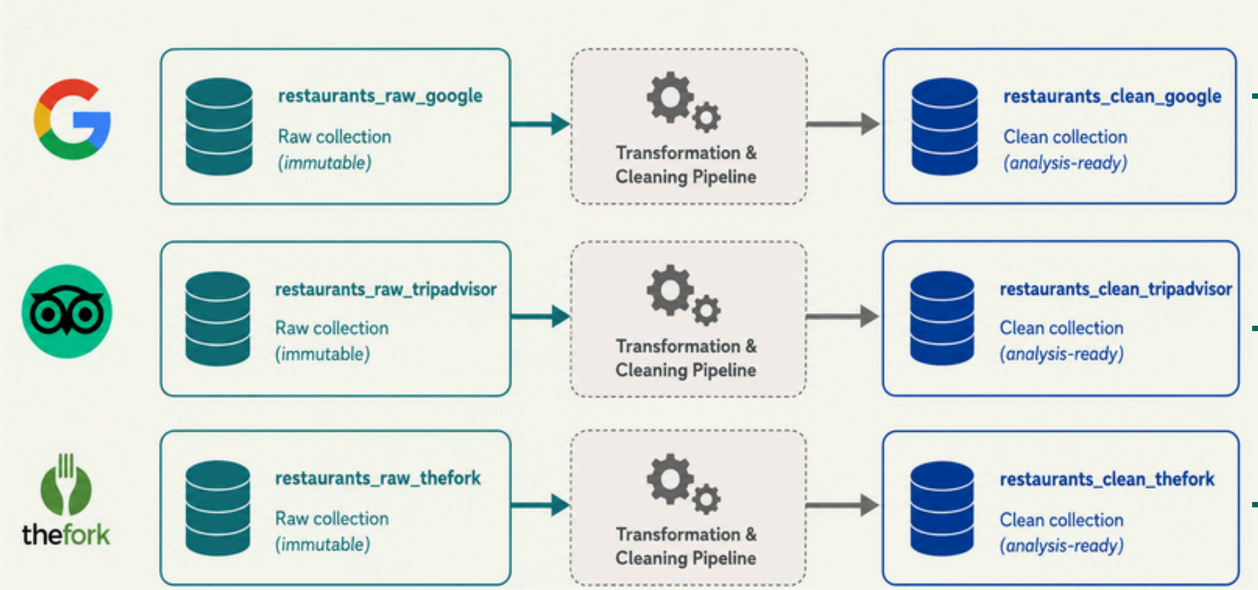
- **TheFork** showed the highest overall data quality and reliability.
- **Tripadvisor** suffered from missing coordinates and low data timeliness due to slow scraping.
- **Google Places** provided the strongest geographic coverage and essential attributes.
- Several data issues were systematic rather than random.
- Contact information was often incomplete and unsuitable for matching.

The analysis allowed us to identify the areas requiring cleaning, enrichment and corrective transformations that are essential during integration.

Data Transformation

Data Transformation Architecture

MongoDB-to-MongoDB

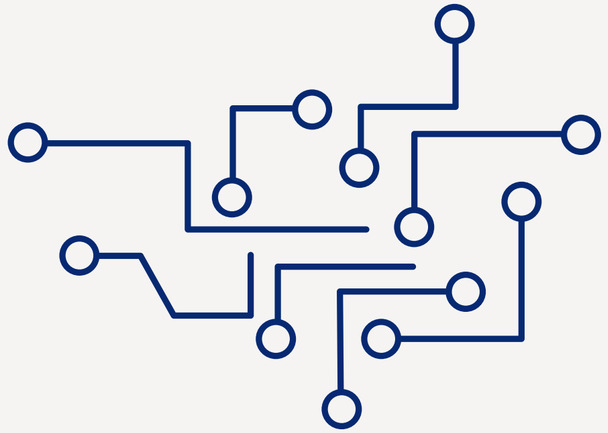


- Removed 22 junk records (not venues)
- Standardized addresses and city names
- Normalized websites and phone numbers
- Extracted structured location attributes

- Parsed ratings and review counts
- Normalized addresses and contact information
- Structured opening hours and cuisine categories
- **Geospatial enrichment** through Nominatim (OpenStreetMap) geocoding → Coverage: 84.0%

- Standardized restaurant names and addresses
- Parsed discounts and average prices
- Normalized cuisine and dietary tags
- Phone and Website were deleted (0% coverage)

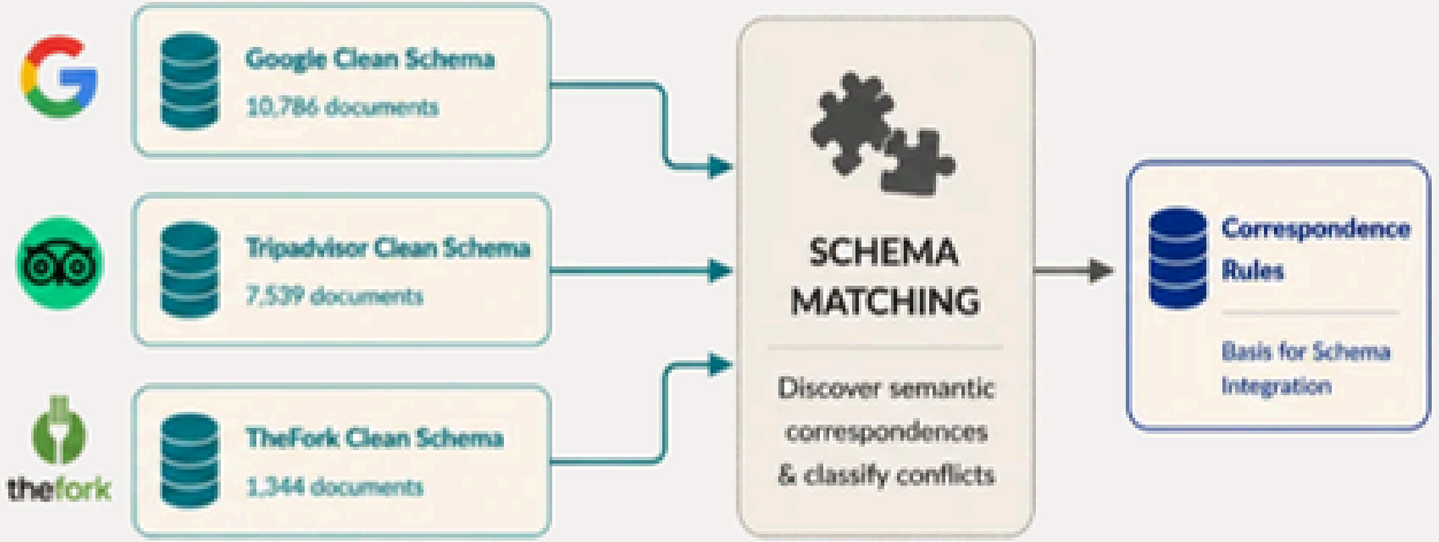
Dataset Integration



Schema Matching

1. Methodology and Framework

Schema matching identifies semantic correspondences between homogenized source schemas before integration.



Matching Relations
(concept level)

- **Equivalence** — Same real-world concept
- **IS-A** — Specialization / subset
- **Overlap** — Partial intersection
- **Disjointness** — Different concepts

Conflict Types
(representation level)

- **Naming Conflicts** — Synonyms or homonyms
- **Structural Conflicts** — Different structure or decomposition
- **Scaling Conflicts** — Different scale or unit (e.g., rating 0–10 vs 1–5)
- **Semantic / Granularity Conflicts** — Different level of detail
- **Coverage Conflicts** — Concept present in one source only

Schema Matching Results

2. Main Correspondences and Integration Decisions

Integrated Concept	Google Clean	Tripadvisor Clean	TheFork Clean	Relation
Restaurant Name	name	restaurant_name	restaurant_name	≡ Same concept, different names.
Locality	locality	city	city	⊑ locality is a more general concept than city.
Cuisines / Food Category	primary_type	cuisines	cuisine_items	⊓ Partially overlapping vocabularies.
Source Venue ID	place_id	ta_location_id	entity_id	⊥ Different identifiers; must not be equated.
Rating	rating (1–5)	rating (1–5)	rating (0–10)	≡ (scale) Same concept, different scale.
Review Count	review_count	total_review	review_count	≡ (naming) Same concept, different names.

≡ Equivalence ⊑ IS-A ⊓ Overlap ⊥ Disjointness

Summary of Inter-Schema Relations Identified

31	15	7	7
Equivalence (≡)	IS-A (⊑)	Overlap (⊓)	Disjointness (⊥)

Examples of Key Conflict Resolution

Rating Scaling

Google	★★★★☆	1–5
Tripadvisor	★★★★☆	1–5
TheFork	★★★☆☆	0–10

→ **Normalized Rating (1–5)**

All ratings converted to a common 1–5 scale.

Price Representation

Google	○○○○	price_level
Tripadvisor	€€ - €€€	price_band
TheFork	€ 32 ○	avg_price_eur

→ **Unified Price Tier**

Different price formats mapped to a unified price tier.

Cuisine Taxonomy

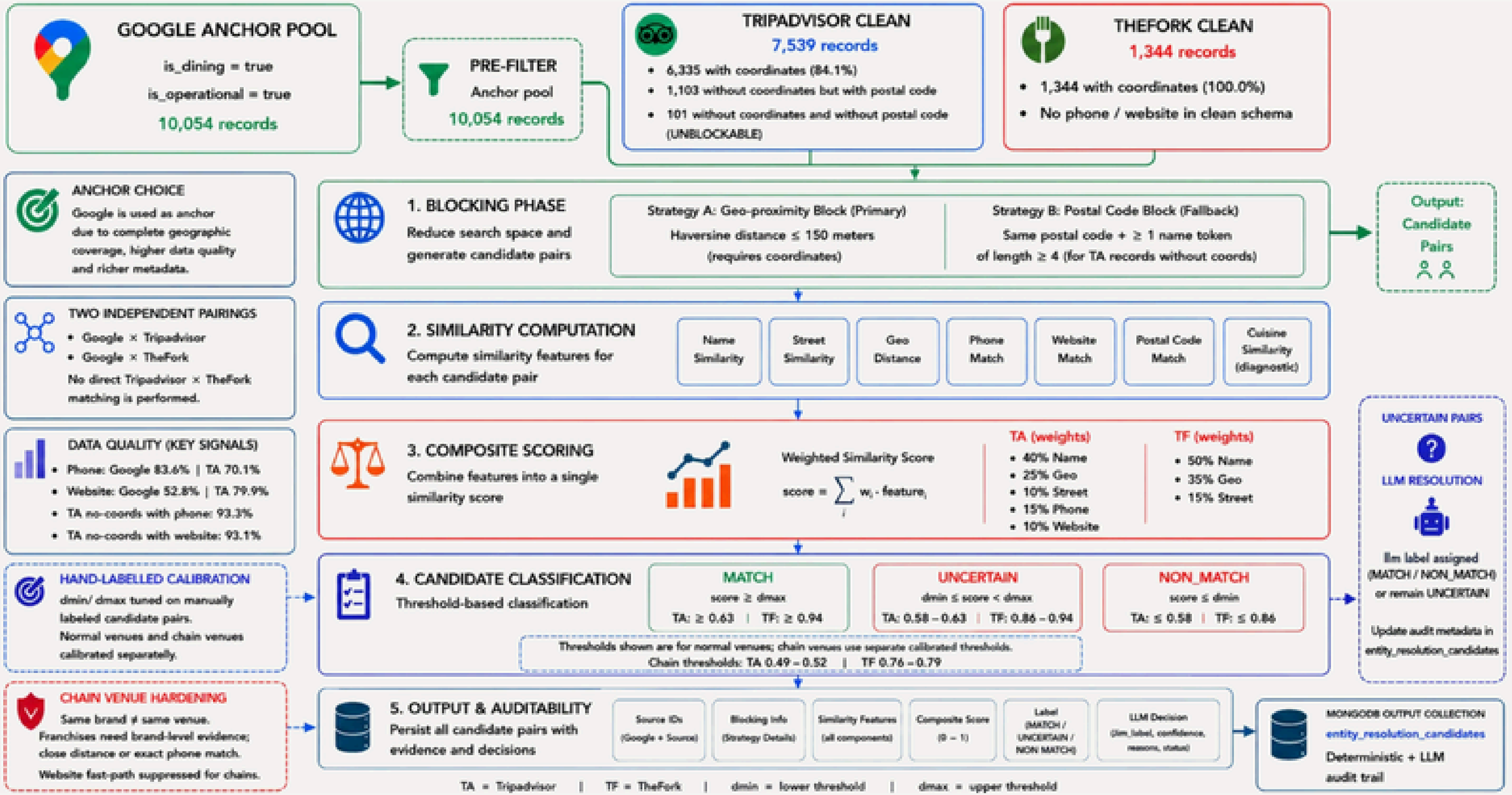
Google	italian_restaurant
Tripadvisor	italiano
TheFork	italiana

→ **Standardized Cuisine**

Cuisine labels standardized to a common taxonomy.

Important Note: Normalized attributes coexist with original source values to support cross-platform analyses.

Entity Resolution Workflow



Quality of Integration

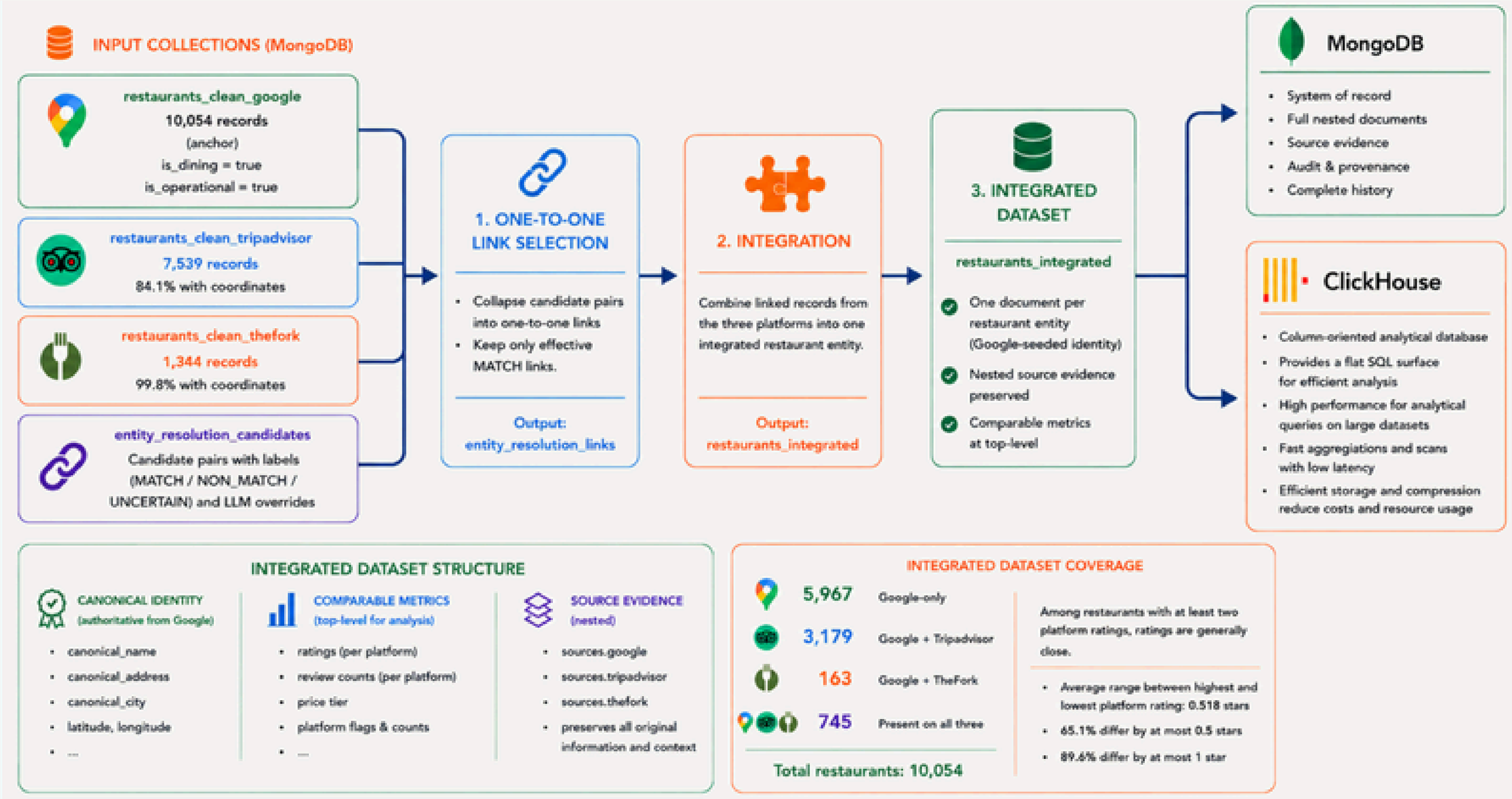
Metric	Out-of-sample	In-calibration
MATCH precision	96.8%	100.0%
Strict MATCH recall	71.7%	92.0%
Kept MATCH recall	100.0%	100.0%
Accuracy	66.0%	96.7%
Uncertain rate	33.0%	3.3%

Metric	Out-of-sample	In-calibration
Geocoding error median	10.1	10.7
Geocoding error p90	40.5	47.1
Geocoding error p95	52.6	83.4
Geocoding error max	92.7	131.5
Within 100 m	100.0%	97.4%
Coordinate coverage	84.0%	84.0%

- Integration results have very high **MATCH** precision (96.8%), so *accepted matches are usually correct*.
- Strict recall is lower (71.7%) because many difficult cases are marked as **UNCERTAIN** rather than forced.
- Since kept **MATCH** recall is 100%, true matches are not lost, only deferred for review.
- Geospatial enrichment went really well with high accuracy, on average ~10m on matched venues.

This is a good result for integration because avoiding false matches is more important than automatically matching everything.

Schema Integration for OLAP Layer



Analytics



RQ1. Are some platforms systematically optimistic/pessimistic?

Platform ratings show a clear systematic gap. Tripadvisor tends to rate restaurants lower, while Google and TheFork are much more closely aligned.

SQL QUERY

```sql
SELECT * FROM (
    SELECT
        'google - tripadvisor' AS pair,
        count(*) AS venues,
        round(avg(google_rating_5 - tripadvisor_rating_5), 3) AS mean_signed_diff
    FROM (INTEGRATED)
    WHERE google_rating_5 IS NOT NULL AND tripadvisor_rating_5 IS NOT NULL

UNION ALL

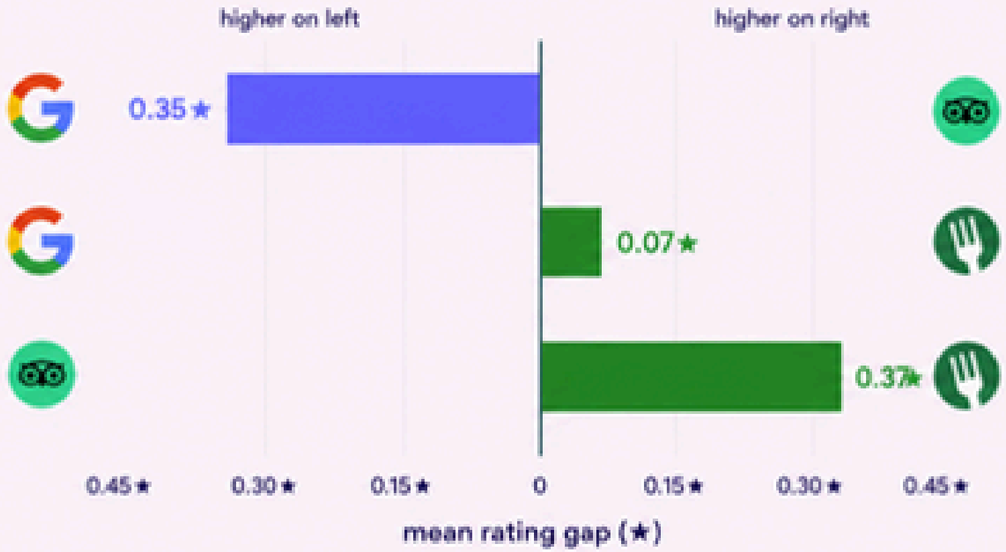
    SELECT
        'google - thefork' AS pair,
        count(*) AS venues,
        round(avg(google_rating_5 - thefork_rating_5), 3) AS mean_signed_diff
    FROM (INTEGRATED)
    WHERE google_rating_5 IS NOT NULL AND thefork_rating_5 IS NOT NULL

UNION ALL

    SELECT
        'tripadvisor - thefork' AS pair,
        count(*) AS venues,
        round(avg(tripadvisor_rating_5 - thefork_rating_5), 3) AS mean_signed_diff
    FROM (INTEGRATED)
    WHERE tripadvisor_rating_5 IS NOT NULL AND thefork_rating_5 IS NOT NULL
) ORDER BY pair;
```

Bias Analysis →

Q1 – Pairwise average rating gap (★)



Bars point toward the higher-rated side, so direction carries the sign. Tripadvisor is lower in both head-to-head checks (−0.35★ and −0.37★); the Google–TheFork gap is only +0.07★, showing they are closely aligned.

MAIN OBSERVATIONS

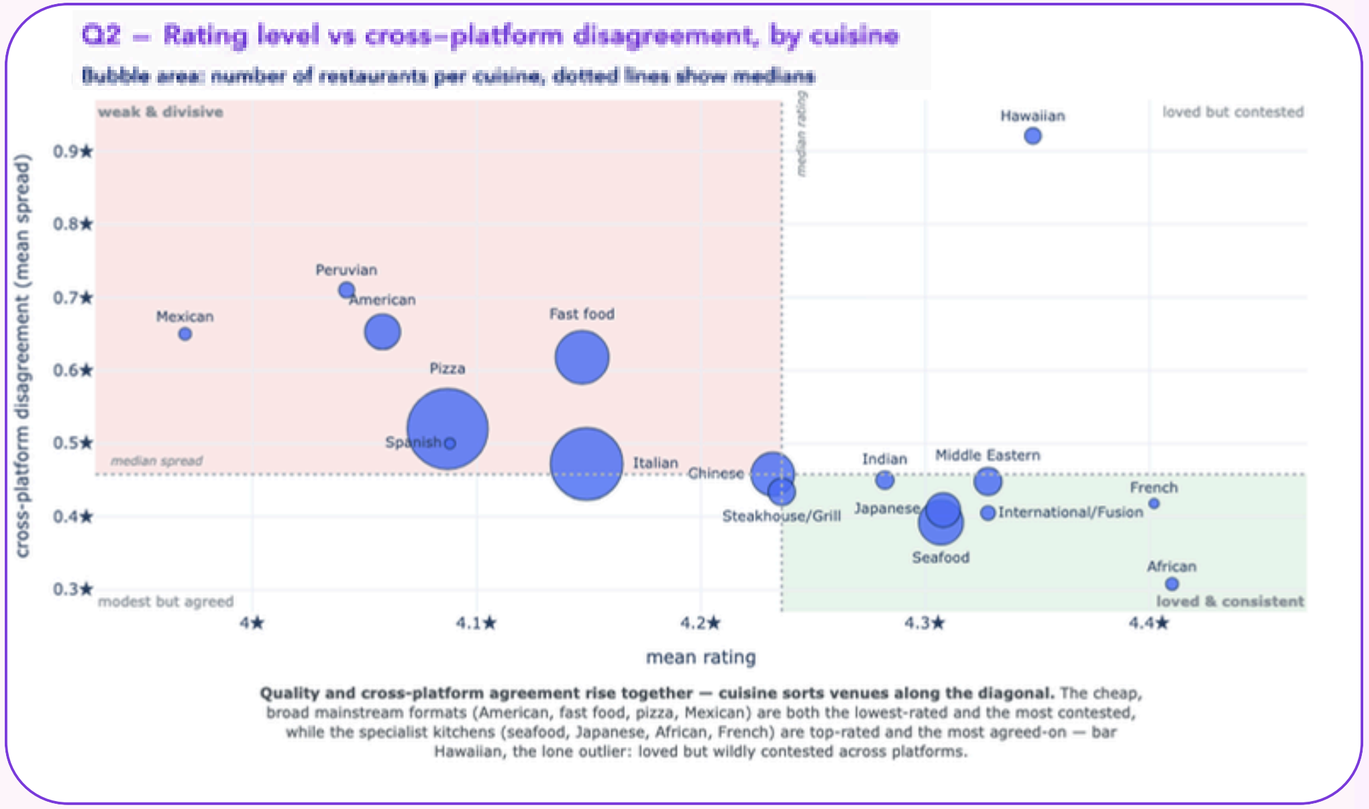
- Google rates restaurants about 0.35★ higher than Tripadvisor.
- TheFork rates restaurants about 0.07★ higher than Google, indicating strong alignment between the two.
- The largest gap is between Tripadvisor and TheFork, at roughly 0.37★.
- Overall, Tripadvisor appears systematically more conservative than the other two platforms.

RQ2. Does rating consistency or level vary by cuisine?

SQL QUERY

```sql
SELECT
        cuisine_primary AS cuisine,
        count(*) AS restaurants,
        round(avg(rating_avg_5), rating_avg_5 IS NOT NULL, 1) AS mean_rating,
        countif(rating_platform_count >= (min_platforms)) AS multi_platform,
        round(multi_platform_count*1.0 / rating_platform_count, 2) AS platform_coverage,
        round(rating_range5 IS NOT NULL, 1) AS mean_rating,
        round(medianif(rating_avg_5, rating_platform_count >= (min_platforms))
                , 1) AS median_rating,
        round(median(consistency(google_review_count, 0) +
            coalesce(tripadvisor_review_count, 0) + coalesce(thefork_review_count, 0))
            /  coalesce(thefork_review_count, 0), 1) AS mean_review,
        round(median(coalesce(google_review_count, 0) +
            coalesce(tripadvisor_review_count, 0) + coalesce(thefork_review_count, 0) *
            coalesce(price_tier, price_tier IS NOT NULL)), 1) AS mean_price,
        avg(price_tier IS NOT NULL) AS priced
FROM (INTEGRATED)
WHERE cuisine_primary NOT IN ('',  'Other')
GROUP BY cuisine
HAVING restaurants >= (min_restaurants)
ORDER BY restaurants DESC
LIMIT 30
```

Cuisine Analysis



Quality and cross-platform agreement rise together — cuisine sorts venues along the diagonal. The cheap, broad mainstream formats (American, fast food, pizza, Mexican) are both the lowest-rated and the most contested, while the specialist kitchens (seafood, Japanese, African, French) are top-rated and the most agreed-on — bar Hawaiian, the lone outlier: loved but wildly contested across platforms.

MAIN OBSERVATION

- Quality and cross-platform agreement rise together: cuisines align along the diagonal.
- Mainstream cuisines (American, Fast Food, Pizza, Mexican) are both the lowest-rated and the most contested.
- Specialist cuisines (Seafood, Japanese, African, French) are top-rated and the most agreed-on.
- Hawaiian cuisine is the lone outlier: highly appreciated but wildly contested across platforms.

RQ3. Does location affect data completeness?

Data completeness is strongly spatially structured.
Central districts exhibit substantially richer metadata than peripheral areas.

SQL QUERY

```
SELECT
      integrated_restaurant_id AS id,
      canonical_name AS name,
      canonical_city AS city,
      canonical_postal_code AS postal_code,
      latitude,
      longitude,
      platform_count,
      has_google, has_tripadvisor, has_thefork,
      google_has_website, google_has_phone,
      tripadvisor_has_website, tripadvisor_has_phone, tripadvisor_has_email,
      website, primary_cuisine,
      google_photo_count, tripadvisor_photo_count, thefork_photo_count,
      google_review_count, tripadvisor_review_count, thefork_review_count,
      rating_range_5
FROM {INTEGRATED}
WHERE canonical_city = 'Milano'
```

Spatial Analysis

Q3 – Distribution of completeness across Milan



MAIN OBSERVATIONS

- Highest completeness in the historical centre (>47%).
- Peaks in Navigli and Isola neighbourhoods.
- Completeness gradually decreases towards the outskirts.
- Outer districts show lower data completeness (≈30%).



THANKS FOR YOUR ATTENTION!